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Development of Climate Change Adaptation Strategies for Rice-wheat Cropping System of Punjab Pakistan

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Theme: Uncertainty and Risk

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Development of Climate Change Adaptation Strategies for Rice-wheat Cropping System of Punjab Pakistan

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Abstract

Climate change is an established fact and its impacts on water, agriculture, health, biodiversity, forest and socio-economic sectors are quite visible around the globe. By virtue of its geographical location, Pakistan is highly vulnerable to the consequent climate change because of global warming. Adaptation to climate change and building resilience among ecosystems and people to respond to climate variability and hazard threats are relatively new concepts. Rice-wheat cropping system zone is the bread basket of Punjab, Pakistan and comprising more than 1 million farm families. There is an urgent need for improved climate modeling and forecasting that can provide a basis for informed decision-making and the implementation of adaptation strategies. For this study crop growth models (DSSAT and APSIM) were calibrated and evaluated on experimental field data to develop the robust genetic coefficients. Models were validated using farmers' field data. Survey data for rice and wheat in Rice-wheat cropping zone of Punjab were collected for this study. An extensive farm survey for 155 farms from the selected five districts Sheikhupura, Nankana Sahib, HafizAbad, Gujranwala, and Sialkot was conducted. Surveyed data (initial condition, crop management, soil characteristic) were used to create the input files in both DSSAT and APSIM. The past climate of the study region, baseline (1981–2010) was analyzed by using the available weather station data and future climate projection were generated by General Circulation Models (GCMs) for mid-century (2040-2069) under RCP 8.5. Five GCMs (CCSM4, GFDLESM2M, HadGEM2-ES, MIROC5, and MPI-ESM-MR) were used for the generation of climate projections. Trade of Analysis model for Multidimensional (TOA-MD) was used for economic analysis. Results of climate change scenarios showed that there would be mean 2.8 °C rise in maximum and 2.2 °C in minimum temperature for mid-century (2040-2069) in Rice-wheat (RW) cropping system of Punjab, Pakistan. In Rice average yield reduction in DSSAT and APSIM was 15.2% and 17.2% while in Wheat average yield reduction by DSSAT and APSIM was 14.1% and 12% during mid-century (2040-2069). A close agreement was observed between farmer and simulated fine rice yield with good statistical indices such as Root Mean Square Error (RMSE) 409 kg ha^{-1} and 440 kg ha^{-1} with d-index (0.80 and 0.78) for DSSAT and APSIM models, respectively. Economic loss of 83% farm household, if

they continue to use current production technology in changed climate. Adaptation strategies using Representative Concentration Pathways (RAP's) were developed to achieve high productivity and meet the need of growing population, it would be required to increase the planting density, fertilizer use and reduce the irrigation amount up to 15% over current with greater potential for promising cultivars. This suggested that due to projected rise in temperature, the cropping seasons will be affected and 5 days earlier transplanting of rice nursery and two weeks earlier in wheat planting over current would be recommended. These strategies have a significant impact in reducing the vulnerabilities of the changing climate with 33% improvement in rice yield to sustain production in Rice-wheat cropping system. Significant reduction in poverty (5-6%) among farm households, if adaptation takes place in this zone.

Keywords

Multi-models' comparison, GCMs, RCPs, RAPs, Economic Modeling

Introduction

Pakistan is the 2nd largest country in South Asia and 36th in the world in terms of area. The total geographical area of Pakistan is 79.6 million ha out of which 22 million ha are used for crop production. Most of the cultivated area is irrigated, encompassing about 19 million ha. The majority of farmers have small land holdings; about 86 percent of the farms have less than 5 ha and only 5 % of the farms are greater than 10 ha (Government of Pakistan, 2010). Punjab is the largest province of Pakistan both with respect to population and agricultural production. Historically the name "Punjab" is derived from two words of the local language i.e. "Punj" meaning Five and "Ab" meaning Water. Punjab encompasses five rivers, and, as a result, it has a well-developed canal system for irrigation

There are two cropping seasons in Pakistan, i.e., Rabi and Kharif. Rabi crops are grown normally from November to April and Kharif crops are grown from May to October. These two seasons determine the agricultural economy of Pakistan. During the Rabi season wheat is dominant; during the growing season 2013-14, it was cultivated on 9.04 million ha in 2013-14 with a total production of 25.3 million tonnes. During the Kharif season rice is also an important crop; it was cultivated on 2.79 million ha in 2013-14 with a production of 6.8 million tonnes. Sustainable production of wheat and rice is crucial for food security in Pakistan as wheat is the staple food of the people and it accounts for almost 10.3% of the total value added in agriculture. Rice is the second major source of food after wheat and accounts for 3.1% of the total value added in agriculture. Historic data show that the area under cultivation and production of rice and wheat was inconsistent. This inconsistency was because of water availability, shift in the

monsoon patterns, and Government policies for support prices of crops and subsidies for inputs (Government of Pakistan, 2014).

Pakistan is located on a large landmass north of the tropic of cancer between the latitude 23° to 38°N and longitude of 60° to 80°E. The climate is defined as continental characterized by extreme variation in temperature, both seasonally and daily. Pakistan is also noted for its cold winters and hot summers. Two-third of its arable land ranges from semi-arid to arid climatic conditions, while a small zone of northeast Punjab adjoining Kashmir experiences sub-humid to humid conditions. The study region lies in the Semi-Arid to sub-humid agro climatic zone, based on the moisture index, (Chaudhry et al., 2004) where the maximum summer monsoon occurs during July and August. The monsoon season from July to September is generally associated with the monsoon depressions that are formed over the Bay of Bengal. They reach Pakistan via India due to their westward motion depending upon their strength. Another cause of summer precipitation is the southwesterly flow of moisture from the Arabian Sea, which is activated with the persistence of a meso-scale low pressure over land. Both phenomena of monsoon and moisture from Arabian Sea reinforce the precipitation process and produce high intensity rainfall. The rice crop mainly depends upon rainfall during monsoon season (July-September). An increase in precipitation by 5% and 15% during September-October could cause decline in rice yield by 5.71% and 15.26%, respectively for the future three decades (Mahmood et al., 2012). A change in local weather conditions due to climate change threatens the productivity of the agricultural sector making it vulnerable both biophysically and economically. Some regions are likely to benefit from climate change while other regions are could be adversely affected. An increase in temperature and changes in rainfall trends and extreme weather events are raising concerns about the prospective impact on water, food and energy. There are five main risks related to climate change in Pakistan, including a rise in sea level, glacial retreats, more and heavier floods, higher temperatures, and an increase in the frequency of droughts. Such risks elevate major challenges for current and future decision-making and have an intricate impact on agriculture, water resources, and urban-rural management, and the overall economy. Keeping in view these challenges the present study was planned with objectives; evaluating the impact of climate change on rice wheat cropping system and development of adaptation strategies,

Methods

Socio-Economic Data

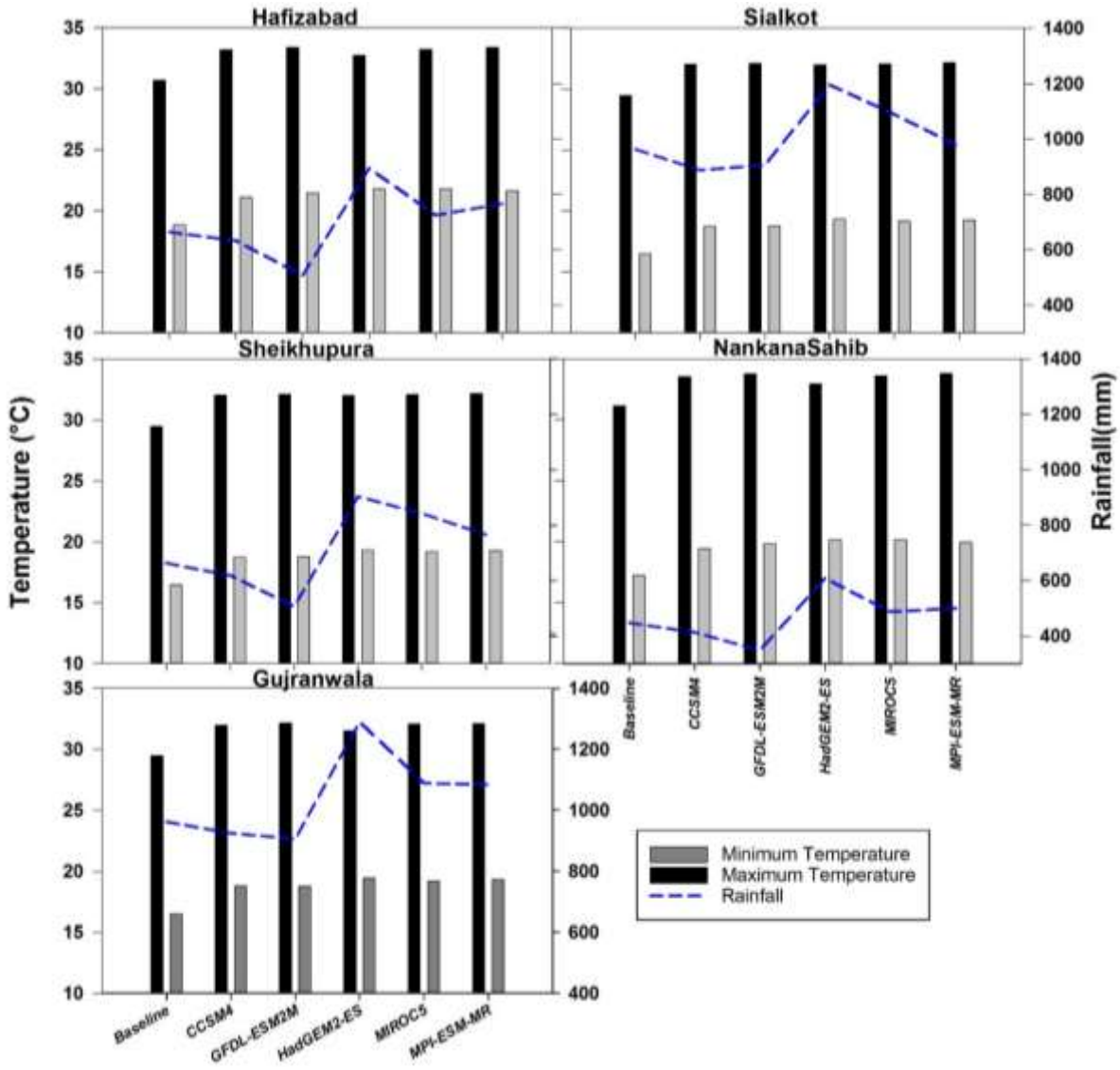
Survey data for rice, wheat and livestock for the Rice-Wheat cropping zone of Punjab were collected for this study. An extensive farm survey for 155 farms from the selected five districts was conducted. Because the population was heterogeneous in nature, a stratified random sampling technique was used. The districts of this study included Sheikhupura, Nankana Sahib, Hafizabad, Gujranwala and Sialkot. Two villages were taken randomly from each district. Each district was taken as a separate stratum, because of its own climatology and topography. From each stratum at least 30 respondents and at least 15 farms from each village were chosen randomly so that the selected sample could be the true representation of the farm population. Missing values, like the management practices (sowing and harvesting dates, non-farm income and other crops etc.) were supplemented from the same farmers later on through personal contact. The economic analysis was conducted on a per farm basis

Climate Change Projections

Delta scenarios (Wilby *et al*., 2004) are based on historical baseline daily weather data; with each day's weather variables perturbed using the changes in climate model outputs for future mid-century RCP8.5 (Moss *et al.,* 2010) versus those same model outputs for the historical time period. The five CMIP5 GCMs i.e., CCSM4, GFDL-ESM2M, HadGEM2-ES, MIROC5 and MPI-ESM-MR (Taylor *et al.,* 2012) used for the generation of climate projections. The selection criterion of these GCMs was based on several facts such as the track record of publications, better performance in monsoon regions and reputation of the model developing institute (Rozenweig et al., 2014).

The baseline (1981-2010) was taken as reference for comparison to assess the change in the projected climatic parameters. The mean maximum, minimum temperatures and precipitation for the RCP8.5 mid-century projections for all stations are presented in Figure 1 along with the baseline. All five selected models performed consistently to project the thermal regime for the future in terms of daily maximum and minimum mean temperatures. Figure 5 shows an increase of about 2°C in temperature during the mid-century (2040-2069) period under the RCP8.5 scenario. However, a heterogeneous behaviour of the models' output has been seen in case of precipitation projections for the selected study area. There is a remarkable difference among the output of the GCMs in the projected values of precipitation for all five districts of Punjab. The HadGEM2-ES was the wettest of all five, projecting a 200 mm increase in precipitation on average in the area of study, while GFDL-ESM2M was the driest showing an expected decrease of about 100 mm for 2040 to 2069. The ensemble mean of five GCMs show a slight increase in annual rainfall (mm) as compared to the baseline.

In order to further examine projected climate change for the mid-century RCP8.5 scenario for the selected stations, the annual cycle of maximum temperature, minimum temperature and precipitation was calculated for each district. There is a clear signal of warming for all five selected districts as the five CMIP5 GCMs show an increase of 2°C to 3°C as compared to the baseline. However, this increase is not uniform for all five CMIP5 GCMs. For example, CCSM4 is showing more warming for the month of April and GFDL-ESM-2M is showing the same for the month of September. There are some exceptional cases like projection of MPI-ESM-MR where the rainfall shows an abrupt increase and decreases as compared to the normal pattern. There is mixed behaviour of GCMs towards projected precipitation as compared to the



baseline. Some CMIP5 GCMs (e.g., HadGEM2-ES and MIROC5) are wet while other GCMs (e.g., CCSM4 and GFDL-ESM-2M) are dry as compared to the precipitation of the baseline period.

Fig.1: Climatology (Tmax, Tmin, and precipitation) for the baseline period (1980-2010) and projections for mid-century RCP 8.5 scenario for (a) Hafizabad, (b) Sialkot, (c) Sheikhupura, (d) Nankana Sahib, and (e) Gujranawala

Crop Modeling

Crop models are being used to evaluate both the spatially and temporally production uncertainties due to crop management and climate scenarios (Hoogenboom, 2000). Crop models have their applications in the fields of research, decision support, education and training (Jones et al., 2003). Crop simulation models have the potential to help understanding the impact of climate change, climate variability and agronomic practices on plant growth and development as the models integrate the soil-plant-atmosphere complex. Two crop growth models were used in this study, including DSSAT and APSIM. Model were first calibrated with experimental data set and evaluated at farmer field level, after that these model were used for climate change impact assessment

Results

On-farm evaluation of crop models

The household surveyed data described in section 3.2 were used to create crop management files for both models. The initial conditions, input levels and soil characteristics of the field that were obtained from the surveys were used to define the input data for both models. Then the models were run with observed weather data of the cropping year i.e. 2010.

DSSAT-Rice

The CSM-CERES-Rice model was evaluated with farmers' field data for 155 fields of rice for five districts of Punjab in the Rice-Wheat cropping zone. There was a good agreement between predicted and observed farmer rice yield with an RMSE of 409 kg ha^{-1}, a d-stat of 0.80 and a bias of 0.94. The CSM-CERES-Rice model simulated rice yield with percent error ranging from -17.6% to 24% and R^2 value of 0.53 as shown in figure 2. The performance of model differed for good and poor management practices of the farmers. The difference between simulated and observed yield was less for those farmers whose management practices were according to recommendations. Planting time, plant population, number of irrigations, irrigation at critical stages, fertilizer application dates, application at crop critical stages, weed management and disease control were better in case of progressive farmers field and model also simulated almost same yield as observed.

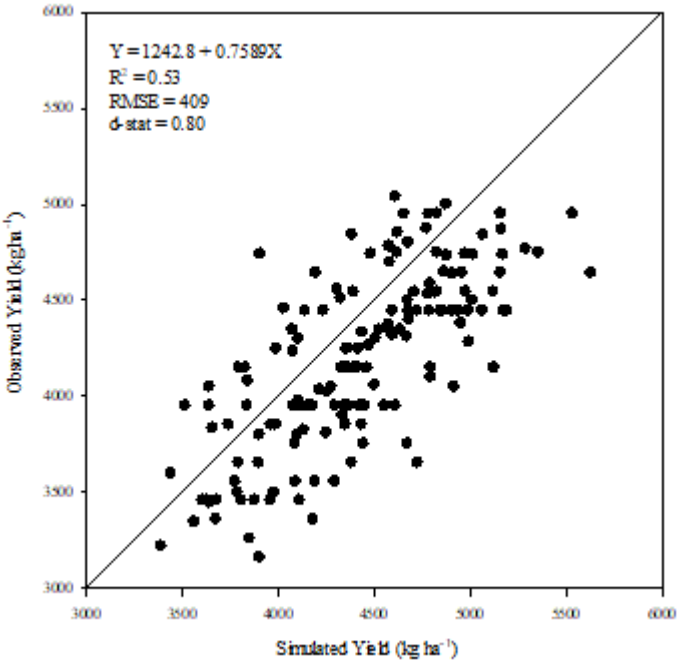


Fig. 2: Relationship between observed and simulated 155 farmers' rice field (using APSIM) in five strata of Rice-Wheat cropping zone of Punjab-Pakistan

APSIM-Rice

The APSIM-Oryza-2000 was evaluated with farmers' field data. Model performed well with an R^2 of 0.44, a RMSE of 440 and a d-stat of 0.78 for the final yield for 155 farmers' fields. The performance of model was different for different farmers. The reason for this difference was the management practices of the farmers and use of different amount of inputs. The difference between simulated and observed yield was less for those farmers who used recommended cultural practices to grow crop. The APSIM had a slightly higher error as compared to the CSM-CERES-Rice model (440 kg ha^{-1} versus 409 kg ha^{-1}) as in figure 3, although the difference was not significant given the uncertainty associated with farmers' management practices and yield as well as the lack of local soil and weather data.

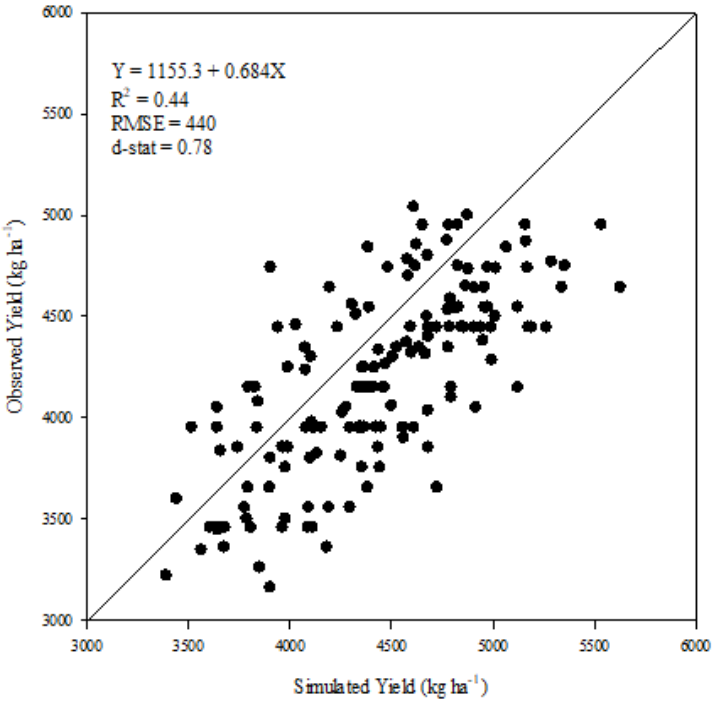


Fig. 3: Relationship between observed and simulated 155 farmers' rice field (using APSIM) in five strata of Rice-Wheat cropping zone of Punjab-Pakistan

DSSAT-Wheat

The CSM-CERES-Wheat model was evaluated with 155 farmers' field to predict wheat yield. A close agreement was observed between the simulated and observed for farmers' yield. When the models were evaluated at the farmers' field, goodness of model (R^2) was 0.64 for CSM-CERES-Wheat between observed and simulated yield of 155 farmers as shown in (Fig 4). Comparison of individual farmer yield showed that DSSAT simulated wheat yield with percent error ranging from -25% to 17% having RMSE 436 kg ha^{-1} and d-statistic of 0.87. Overall, DSSAT model predicted yield with error 1.6% and bias was calculated as 0.98. The difference between simulated and observed yield was less for those farmers who were growing wheat according to agriculture department recommendations. Good cultural practices by progressive farmers were the key factors in better simulation of model.

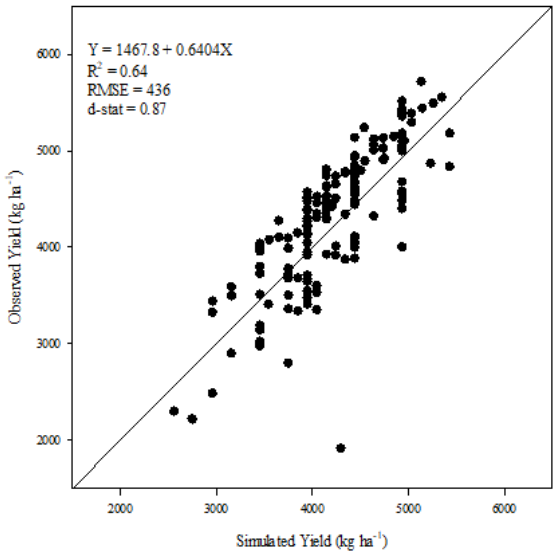


Fig. 4: Relationship between observed and simulated 155 farmers wheat field (using DSSAT) in five strata of Rice-Wheat cropping zone of Punjab-Pakistan

APSIM-Wheat

The APSIM-Wheat was evaluated with farmers' field data of 155 farmers of five regions. The relationship was drawn between farmers' observed yield and simulated yield of wheat by APSIM. Model performed well with an R^2 of 0.37, a RMSE of 592 and a d-stat of 0.88 as shown in figure 5. APSIM-wheat simulated yield with more error as compare to DSSAT as values for RMSE and d-stat were slightly higher as compared to DSSAT. So far as the value of R^2 was concerned, it was less than CSM-CERES-Wheat which showed that the performance of CSM-CERES-Wheat was slightly better as compared to APSIM. However, the wheat data from the farmers' fields had the same uncertainty as the rice data.

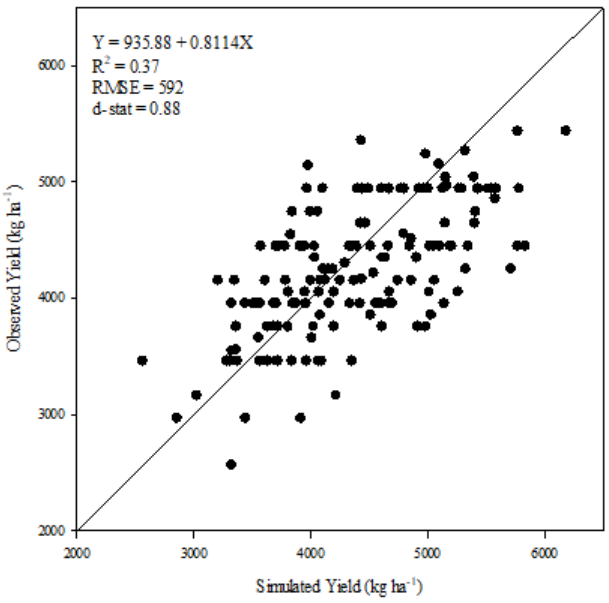


Fig. 5: Relationship between observed and simulated 155 farmers wheat field (using APSIM) in five strata of Rice-Wheat cropping zone of Punjab-Pakistan

Economic Analysis

The Tradeoff Analysis Model for Multi-Dimensional Impact Assessment (TOA-MD) Version 6.0.1 Beta (Antle and Valdivia, 2014) was used to determine the impact of climate change on socio-economic indicators, TOA-MD is a parsimonious and generic model that has been used for the analysis of technology adoption and ecosystem services. It simulates technology adoption and impact in a population of heterogeneous farms households. Farmers are assumed to be economically rational and thus choose among management systems based on expected economic returns. The model uses data on the spatial variability in economic returns to represent heterogeneity in the farm population. One of the important implications of this model is that incomplete adoption of a new technology can simply be due to heterogeneity in the farming conditions such as soils, climate, water, costs, and the farm household's characteristics in technology adoption analysis.

Most of the literature attributes incomplete adoption to attitudes such as risk aversion or constraints like access to the technology or finance that are typically difficult to observe and quantify (Suri, 2011). While these factors may indeed contribute to low adoption rates in some cases, in several cases observable heterogeneity in bio-physical and economic characteristics of farms can be sufficient to explain low adoption rates (Antle *et al*., 2005). In climate change assessment, the TOA-MD model implies that all farms are not affected in the same way, some

farms lose and some farms gain from climate change. Similarly, some farms may be willing to adopt technologies that facilitate adaptation to climate change, while others will not. The TOA-MD model allows researchers to simulate the impacts of the full range of adoption rates from 0 to 100 percent. In this study farmers are presented with a simple binary choice: they can operate with a current or base production system, referred to as "System 1", or they can switch to an alternative system referred to as "System 2".

Under the climate change analysis, it is necessary to distinguish between three factors affecting the expected value of a production system: the production methods used, referred to here as the technology, physical environment in which the system is operated, i.e., the climate, economic and social environment in which the system is operated, i.e., the socio-economic setting that has been referred as Representative Agricultural Pathways (RAPs) (Rosenzweig *et al.,* 2013b). These RAPs are basically qualitative storylines that can be translated into model parameters such as farm and household size, prices, costs of production and policy. A major challenge in scenario design for climate impact assessment is the dimensionality of the analysis. Farmers are initially operating a base technology with a base climate known as 'System 1' while 'System 2' is defined as the case where farmers continue using the base technology under perturbed climate. There is also a 'System 3' which is defined as an adapted technology used to withstand the adverse impacts of perturbed climate.

The analysis was conducted for each of 5 GCMs discussed previously. Their overall impacts were observed in each scenario for both the DSSAT and APSIM crop model outcomes and for livestock. Three Core Questions were answered through the analyses. In Core Question 1, sensitivity of current agricultural production system to climate change was assessed. Impact of climate change on future agricultural production was addressed in Core Question 2. Core Question 3 was to find out benefits of potential adaptations to climate changes (Rosenzweig et al*.,* 2013b). First, climate change impact assessments (CC-IA) were made without using the RAPs to find out the sensitivity of current agricultural production system to climate change (Core Question 1). After applying the RAPs the analyses were again carried out for evaluating the impacts of climate change on future production system (Core Question 2). Then analysis was made for future adapted production system by using RAPs and adaptations to address Core Question 3. The outputs of the rice and wheat simulations of DSSAT and APSIM were used as inputs of TOA-MD.

Data used for Economic Analysis

The data used for TOA-MD analysis for the base system is given in the form of the average for all selected strata. The average sampled farm area per stratum was 176.5 hectare. While the

average yields of rice and wheat crops were 18,376 kg and 18,921 kg per farm, respectively. The output price per kg was Rs. 43 for rice, and Rs. 29 for wheat. Variable cost of production for rice was Rs. 533,981 per farm and for wheat was Rs. 338,348 per farm. Mean net farm returns for rice and wheat crops were Rs. 257,513 and Rs. 204,908 per farm, respectively.

Integrated Assessment Results

Sensitivity of Current Agricultural Production Systems to Climate Change

The changing climate is worsening the current vulnerabilities of the small farmers, especially those who depend on semi-subsistence agriculture for their livelihood (Nelson *et al*., (2009) cited in Claessens *et al*., (2012)). The first analysis is based on the 'current technology, perturbed climate' scenario for mid of the century time period i.e. 2040–2069. In the first step of economic analysis, effects of future climate and current production system were analysed for productivity and socio-economic indicators. For each GCM, gains, losses and net impacts for Core Question 1 are presented in Figure 6.

It is clear from the analysis that the observed mean production of wheat is 18,915 kg per farm across the study area. The reduction in mean yield for DSSAT ranged from 6 to 19 % while for APSIM it ranged from 10 to 12 % for all 5 GCMs. In the case of rice, the production was 18,349 kg per farm for the study area. The reduction in mean yield for DSSAT ranged from 8 to 30 % while for APSIM it ranged from 14 to 19 %. In case of livestock, observed mean production of milk was 3,267 litres per farm for all GCMs with a reduction in production of about 12% for Core Question 1 which is about finding the sensitivity of current agricultural production systems to climate change. Income from the sale of animals was added in the non-agricultural income in the farm sheet for the economic analysis since dairy farming was one of important activity in the study area. The same pattern for livestock income was found for Core Questions 2 and 3. The analyses of these Core Questions assessed the impacts of climate change on future agricultural production systems and the benefits of climate change adaptations.

Depending upon the distribution of costs and benefits across the area, the analysis with showed that number of losers ranged from 69 to 83 percent for DSSAT and from 72 to 76 percent for APSIM, due to perturbed climate. Percentage gains and losses for DSSAT ranged from 13 to 15 % and 23 to 30 %, respectively. While for the case of APSIM percentage gains and losses ranged from 14 to 15 % and from 25 to 27 %, respectively. The observed net returns without climate change for both DSSAT and APSIM were Rs. 0.54 million per farm. However, with the adverse climate change, the net returns varied from Rs. 0.42 to 0.48 million per farm for DSSAT and from Rs. 0.45 to 0.47 million per farm for APSIM for all five GCMs. Observed

per-capita income without climate change would be about Rs. 70 thousand per person per year for all cases. While with climate change it would reduce and would vary between Rs. 56,000 to 63,000 per person per year for DSSAT and Rs. 59,000 to 61,000 per person per year for APSIM, respectively. Without climate change poverty rate would be 29 % while in case of climate change it would increase and show a range from 33 to 38 % for DSSAT and from 35 to 37 % for APSIM, respectively.

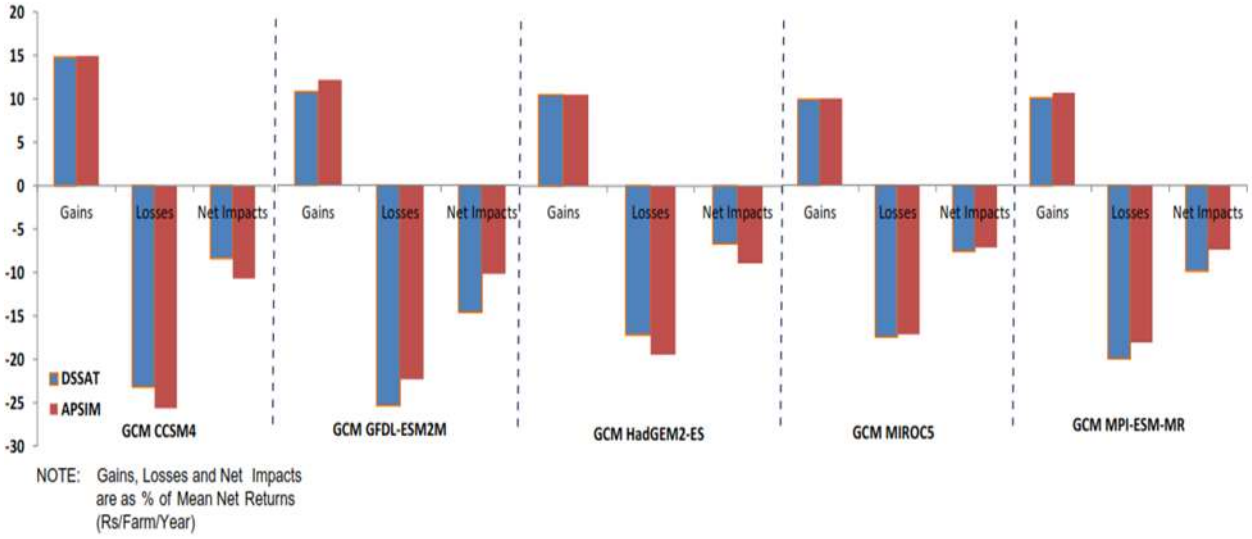


Fig. 6: Distribution of gainers and losers (percentage) for 5 GCMs using DSSAT and APSIM crop models for Rice-Wheat cropping system of Punjab (answering Core Question 1)

The Impacts of Climate Change on Future Agricultural Production Systems

The second stage of the TOA-MD analysis was to compare the 'System 1' i.e. future production system and current climate with 'System 2' i.e. future production system and future climate in mid-century (2040–2069) for Rice-Wheat cropping system using 5 GCMs. These analyses were conducted to evaluate the isolated role of climate impacts for the selected cropping system using adaptation scenarios. The International Model for Policy Analysis of Agricultural Commodities and Trade (IMPACT) (Nelson *et al*, 2010) was used for price and yield trend factors for both wheat and rice. The IMPACT model data were used with regional RAPs. These trends were in real terms based on 2005 in USD in order to make the regional RAPs consistent with the global RAPs. The yields trends based on IMPACT for rice and wheat were 139 and 126 percent, respectively. Price trends for wheat and rice crops used for without climate change case were 127 and 119 percent, and for with-climate change case these values were 133 and 125 percent, respectively. In case of cost trend factor, these values for wheat and

rice were 139 and 125 percent, respectively. Livestock was trended by using the regional RAPs developed for the Rice-Wheat cropping system of Punjab.

Gains, losses and the net impacts for each GCM were determined to find out the impacts of climate change on future agricultural production systems (Figure 7). The projected wheat production is 26,473 kg per farm for all case of Core Question 2 (Table 8). The reduction in wheat yield ranged from 6.2 to 19 % for DSSAT and from 10.6 to 12.3 % for APSIM. In case of the rice crop, the projected production was 25,073 kg per farm for all analysis of Core Question 2 (Table 8). The reduction in rice yield ranged from 8 to 30 % for DSSAT, while for APSIM it ranged from 14 to 19 %. In case of livestock, projected mean milk production was 3, 267 litres per farm for all analysis and the reduction in production was 12 % for all case of Core Question 2.

The percentage of farmers who would lose their farm income ranged from 57 to 70 % for the DSSAT crop model simulations and from 61 to 71 % for the APSIM crop simulations for all 5 GCMs. For DSSAT the percentage gains would be 11 to 13 % and percentage loss would be from 16 to 22%, respectively. For APSIM the percentage gains would be 10 to 15% and percentage loss would be from 17 to 19%, respectively. Projected net returns without climate change for all case of DSSAT and APSIM would be up to Rs. 1.14 million per farm per year for all five GCMs. Projected net returns with climate change ranged from Rs. 0.97 to 1.1 million per farm for DSSAT and from Rs. 1.04 to 1.08 million per farm for APSIM. The projected per-capita income without climate change would be Rs. 103,000 per person per year for all GCMs and both crop models. However, under climate change, per capita income ranged from Rs. 89,000 to 100,000 per person for DSSAT crop simulations and from Rs. 93,000 to 97,000 for the APSIM simulations. Without climate change poverty rate would be 16 % for both crop models, while under climate change poverty rates would slightly increase and range from 17 to 19 % for DSSAT and from 18 to 19 % for APSIM.

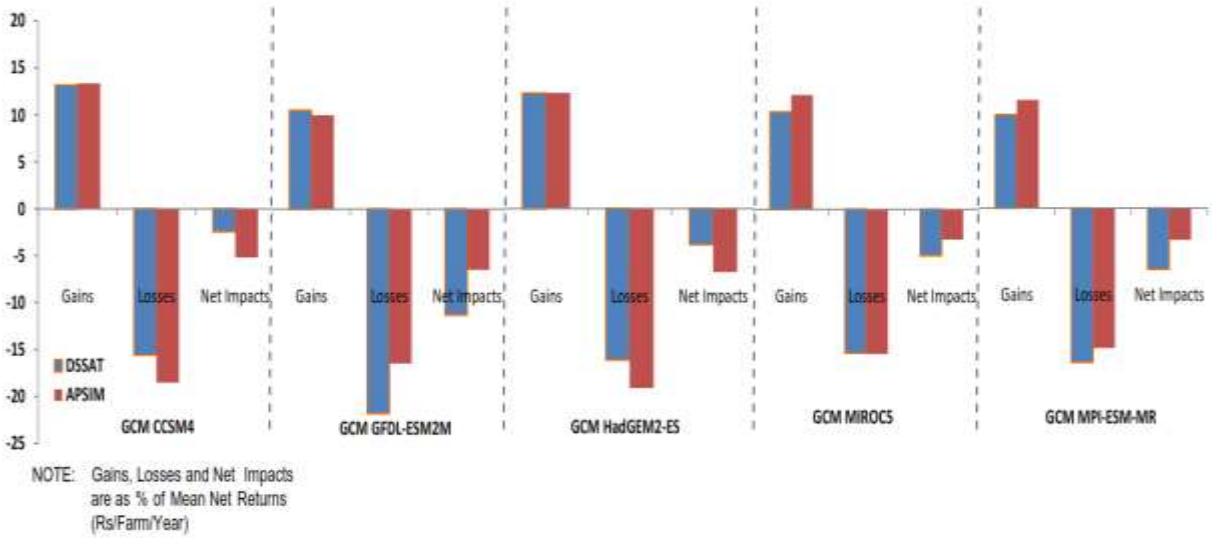


Fig. 7: Aggregated gains and losses with all 5 GCMs for DSSAT and APSIM for Rice-Wheat Cropping System of Punjab-Pakistan (answering Core Question 2)

5.3 Evaluation of Potential Adaptation Strategies

Finally the impact of adaptation strategies on the distribution of potential adopters and non-adopters were analyzed using TOA-MD to answer the Core Question 3. For analysis, in 'System 1' RAPs were included and in 'System 2', RAPs and adaptation package were included for Rice-Wheat system based on the crop model simulations of DSSAT and APSIM. This analysis was basically for future, in which non-adopted and adopted production systems were compared. It is clear that after implementing the adaptations strategies, the vulnerabilities of future cropping system to climatic change have been reduced to a large extent and these adaptations have a significant impact on the Rice-Wheat cropping zone of Punjab.

Table 1: Quantification of scenarios for crop models (DSSAT and APSIM) for rice and wheat crops on the basis of adaptation package in Rice–Wheat cropping zone of Punjab

Serial No.	Variable	Direction of Change	%age Change	
			Rice	Wheat
1	Nitrogen/Hectare (Kg)	Increase	15	25
2	Sowing Density (Plant/m^2)	Increase	15	30
3	Irrigation	Decrease	15	25
4	Sowing Dates	Decrease	5 Days	15 Days

For each GCM, the percentage of adopters for Core Question 3, results are presented in (Figure 8). In the Core Question 3, the benefits of climate change adaptations were assessed. The

projected production of wheat would vary from 21,583 to 24,819 kg per farm for DSSAT and 23,166 to 23,657 kg per farm for APSIM across the study area in adaptation scenario analyses (Table 9). Mean change in wheat yield ranged from 10.2 to 22.6% for DSSAT and from 30.6 to 39.2% for APSIM for all five GCMs. In case of rice, projected production without adaptation ranged from 16,126 to 21,427 kg per farm for DSSAT and from 18,560 to 20,401 kg per farm for APSIM. Mean change in yield for DSSAT ranged from 9.9 to 20.1%, while for APSIM it ranged from 1.7 to 24%. In case of livestock, projected mean milk production without adaptation was 3,593 litres per farm for all GCMs and its mean yield increase was 42% for all cases.

Table 2: Quantification of scenarios for TOA-MD for Rice–Wheat cropping system of Punjab

Serial No.	Variable	Direction of Change	%age Change	
			Rice	Wheat
1	Average House Hold Persons	Increase	40	40
2	Non-Agricultural Income	Increase	40	40
3	Price of Output	Increase	65	70
4	Variable Production Cost	Increase	55	50

The percentage adopters for the DSSAT crop simulations ranged from 69 to 74 % and for APSIM from 70 to 81%. For DSSAT, the projected net returns without adaptation ranged from Rs. 0.97 to 1.1 million per farm per year and with adaptation ranged from Rs. 1.17 to 1.32 million per farm per year. For APSIM the projected net returns without adaptation ranged from Rs. 1.04 to 1.08 million per farm and with adaptation it ranged from Rs. 1.28 to 1.39 million per farm per year. For DSSAT projected per-capita income without adaptation would be from Rs. 89 to 100 thousand and with adaptation it would be form Rs. 107 to 118 thousand per person per annum. In case of APSIM projected per-capita income for without adaptation cases would be between Rs. 93 to 97 thousand and with adaptation it would range from Rs. 114 to 125 thousand per person per annum. Without adaptation poverty would range from 17 to 19 % for the DSSAT crop simulations and from 18 to 19 % for the APSIM crop simulations, while, with adaptation the poverty rate would slightly decrease to 15 to 17 % for both crop simulation models.

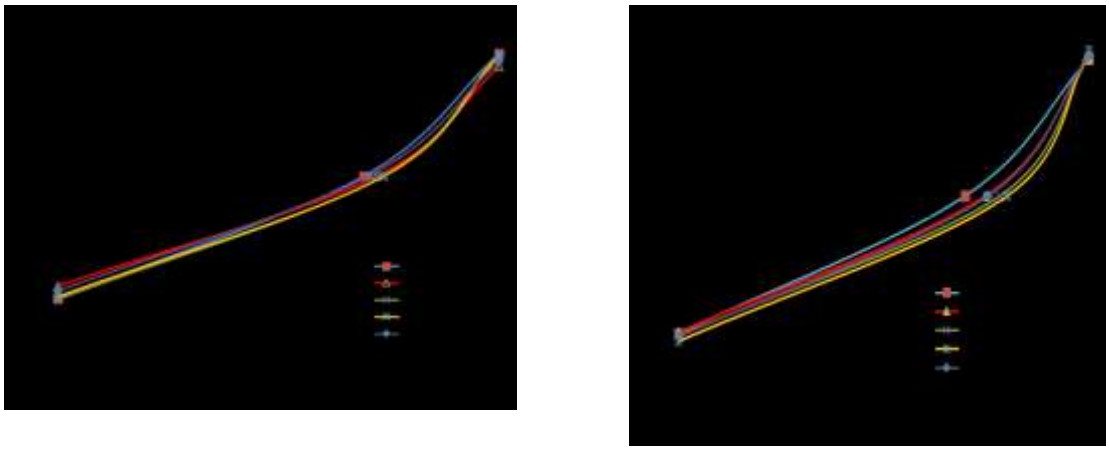


Fig. 8: Adaptation curve showing distribution of adopters and non-adopters for all 5 GCMs for both crop models for Rice-Wheat system of Punjab

Conclusion

Pakistan is exposed to five major risks related to climate change, inclduing sea level rise, glacial retreats, floods, higher average temperature and an increase in the frequency of droughts. Its economy is dependent upon agriculture and these climatic risks are major threats for sustainability of agricultural systems. Any negative impact would be devastating not only for the socio-economic conditions of the rural population but also jeopardizing the food security of the country. It is therefore, important to quantify the impact of climate change in agriculture using an interdisciplinary approach.

The findings of this study showed that the performance of the crop simulations DSSAT and APSIM for simulating rice and wheat yield was good. For rice, DSSAT simulated yield of 155 farms with an R^2 of 0.52, a RMSE of 425 kg ha^{-1} and a d-stat of 0.78, while for APSIM the R^2 was 0.42, the RMSE was 440 kg ha^{-1} and the d-stat was 0.78. For wheat, DSSAT simulated yield with an R^2 of 0.64, a RMSE of 436 kg ha^{-1} and the d-stat was 0.87, while for APSIM the R^2 was 0.37, a RMSE of 592 kg ha^{-1} and a d-stat of 0.88, respectively. Due to climate change, there is a mean reduction of 15.2% and 17.2% in rice yield by DSSAT and APSIM, respectively. In wheat, mean yield reduction was 14.0% in DSSAT and 13.76% in APSIM with five GCMs, respectively.

By using TOA-MD the future socio-economic indicators were estimated to observe the impact of climate change on current and future agricultural production systems for the mid-century scenarios (2040-69). It was estimated that if the current production technology prevails in future, there would be about 69 to 83% losers due to perturbed climate. Poverty rate in changing climate would be between 33 to 38%. Quantification of climate change impact on future

agricultural production systems depicted that there would be 57 to 71% losers and poverty would range between 17 to 19%. In the light of above results, it is clear that climate change could negatively impact the future crop and livestock production and farm returns. However, the adaptation strategies could have a significant positive impact in reducing the vulnerabilities due to climate change. Potential adopters of the adaptation strategies ranged from 69 to 81% and poverty could reduce from 15 to 17% after the implementation of the adaptation strategies.

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